1. A point charge of 20 nC is fixed at the origin of x-y plane. The work done in taking a second point charge of – 50 nC from point (4 cm, 0) to point (0, 5 cm) is :

(A) 45 μJ

(B) – 45 μJ

(C) 90 μJ

(D) – 90 μJ

1. A point charge of 20 nC is fixed at the origin of x-y plane. The work done in taking a second point charge of – 50 nC from point (4 cm, 0) to point (0, 5 cm) is :

(A) 45 μJ (B) – 45 μJ

(C) 90 μJ (D) – 90 μJ

CBSE 25 C

(A) 45 μJ

1. A point charge of 10 nC is fixed at the origin of x-y plane. A second point charge of – 20 nC is moved from point (5 cm, 0) to point (0, 5 cm). The change in the potential energy of the system of the two charges is :

(A) 36 μJ (B) – 36 μJ

(C) 72 μJ (D) 0

CBSE 25 C

1. A point charge of 10 nC is fixed at the origin of x-y plane. A second point charge of – 20 nC is moved from point (5 cm, 0) to point (0, 5 cm). The change in the potential energy of the system of the two charges is :

(A) 36 μJ (B) – 36 μJ

(C) 72 μJ (D) 0

CBSE 25 C

(D) Zero

(ii) A dipole is aligned parallel to the electric field. Find the work done in rotating it through an angle (I) 180°, and (II) 90°.

CBSE 25 C

(ii) A dipole is aligned parallel to the electric field. Find the work done in rotating it through an angle (I) 180°, and (II) 90°.

CBSE 25 C

(ii) $W = U_2 - U_1 = -pE\cos 180° - (-pE\cos 0°)$	
$= pE + pE = 2pE$	½
$W = -pE\cos 90° - (-pE\cos 0°)$	½
$= 0 + pE = pE$	½
	½

28. Two point charges of $-5\ \mu C$ and $2\ \mu C$ are located in free space at $(-4\ \text{cm}, 0)$ and $(6\ \text{cm}, 0)$ respectively.

(a) Calculate the amount of work done to separate the two charges at infinite distance.

(b) If this system of charges was initially kept in an electric field $E = \dfrac{A}{r^2}$, where $A = 8 \times 10^4\ \text{N C}^{-1}\ \text{m}^2$, calculate the electrostatic potential energy of the system. CBSE 25 *3*

(a) $U=\dfrac{kq_1q_2}{r}$	½
$=\dfrac{9\times10^9\times\left(-5\times10^{-6}\right)\times\left(2\times10^{-6}\right)}{10\times10^{-2}}$	
$=-0.9\,J$	
W = U(infinity) – U(r) = 0.9J	½
(b) $E=\dfrac{-dV}{dr}$	
$dV=\dfrac{-A}{r^2}dr$	½
$V=\dfrac{A}{r}$	
$U=q_1\,V_1+q_2\,V_2+\dfrac{kq_1q_2}{r}$	½
$=\left(-5\times10^{-6}\right)\left(\dfrac{8\times10^4}{4\times10^{-2}}\right)+\left(2\times10^{-6}\right)\left(\dfrac{8\times10^4}{6\times10^{-2}}\right)-0.9$	½
$=-10+2.67-0.9$	
$=-8.24\,J$	½

31. (a) (A) Two points charges q_1 and q_2 in air are located at $\vec{r}_1$ and $\vec{r}_2$ respectively in a region of uniform external field $\vec{E}$. Obtain an expression for the electrostatic potential energy of the system.

(B) A potential difference of 480 V is applied between two large horizontal plates, kept one above the other, 3.0 cm apart. A charged particle of mass 4×10^{-11} kg, when released at rest between the plates, remains stationary in the region. If the upper plate is at positive potential, find

(i) nature of the charge on the particle

(ii) the magnitude of the charge on the particle. **5**

CBSE 25

(a) Expression for potential energy.	
Work done in bringing the charge q_1 ,from ∞ to r_1 , $W_1 = q_1 V(\vec{r}_1)$	½
Work done in bringing the charge q_2 , from ∞ to r_2 is not only against the external field but also against the field due to q_1	½
$$W_2 = q_2 V(\vec{r}_2) + \frac{q_1 q_2}{4\pi\varepsilon_o \vec{r}_{12}}$$	1
Where r_{12} is the distance between q_1 and q_2	
Thus , potential energy of the system = total work done in assembling the system	½
$$U = q_1 V(\vec{r}_1) + q_2 V(\vec{r}_2) + \frac{q_1 q_2}{4\pi\varepsilon_o \vec{r}_{12}}$$	½
(B) (i) As charge particle at rest which emplies that electrostatic force balances the gravitational force.	
So, nature of charge is negative.	1
(ii) $$mg = qE = q\frac{V}{l}$$	½
$$q = \frac{4\times10^{-11}\times10\times3\times10^{-2}}{480}$$	
$$q = 2.5\times10^{-14}\ \text{C}$$	½

3. Four point charges Q each, are held at the four corners of a square of side l. The amount of work done in bringing a charge Q from infinity to the centre of the square will be :

(A) $\dfrac{Q^2}{\pi \varepsilon_0 l}$

(B) $\dfrac{\sqrt{2}\,Q^2}{\pi \varepsilon_0 l}$

(C) $\dfrac{Q^2}{2\pi \varepsilon_0 l}$

(D) Zero

CBSE 25

3. Four point charges Q each, are held at the four corners of a square of side l. The amount of work done in bringing a charge Q from infinity to the centre of the square will be :

(A) $\frac{Q^2}{\pi \varepsilon_0 l}$

(B) $\frac{\sqrt{2} Q^2}{\pi \varepsilon_0 l}$

(C) $\frac{Q^2}{2\pi \varepsilon_0 l}$

(D) Zero

CBSE 25

(B) $\frac{\sqrt{2}Q^2}{\pi\varepsilon_0 l}$

3. The electric field at a point in a region is given by $\vec{E} = \alpha \frac{\vec{r}}{|\vec{r}|^3}$, where α is a constant and r is the distance of the point from the origin. The magnitude of potential of the point is :

(A) $\frac{\alpha}{r}$

(B) $\frac{\alpha r^2}{2}$

(C) $\frac{\alpha}{2 r^2}$

(D) $-\frac{\alpha}{r}$

CBSE 25

3. The electric field at a point in a region is given by $\vec{E} = \alpha \frac{\vec{r}}{|\vec{r}|^3}$, where α is a constant and r is the distance of the point from the origin. The magnitude of potential of the point is :

(A) $\frac{\alpha}{r}$ (B) $\frac{\alpha r^2}{2}$

(C) $\frac{\alpha}{2r^2}$ (D) $-\frac{\alpha}{r}$

CBSE 25

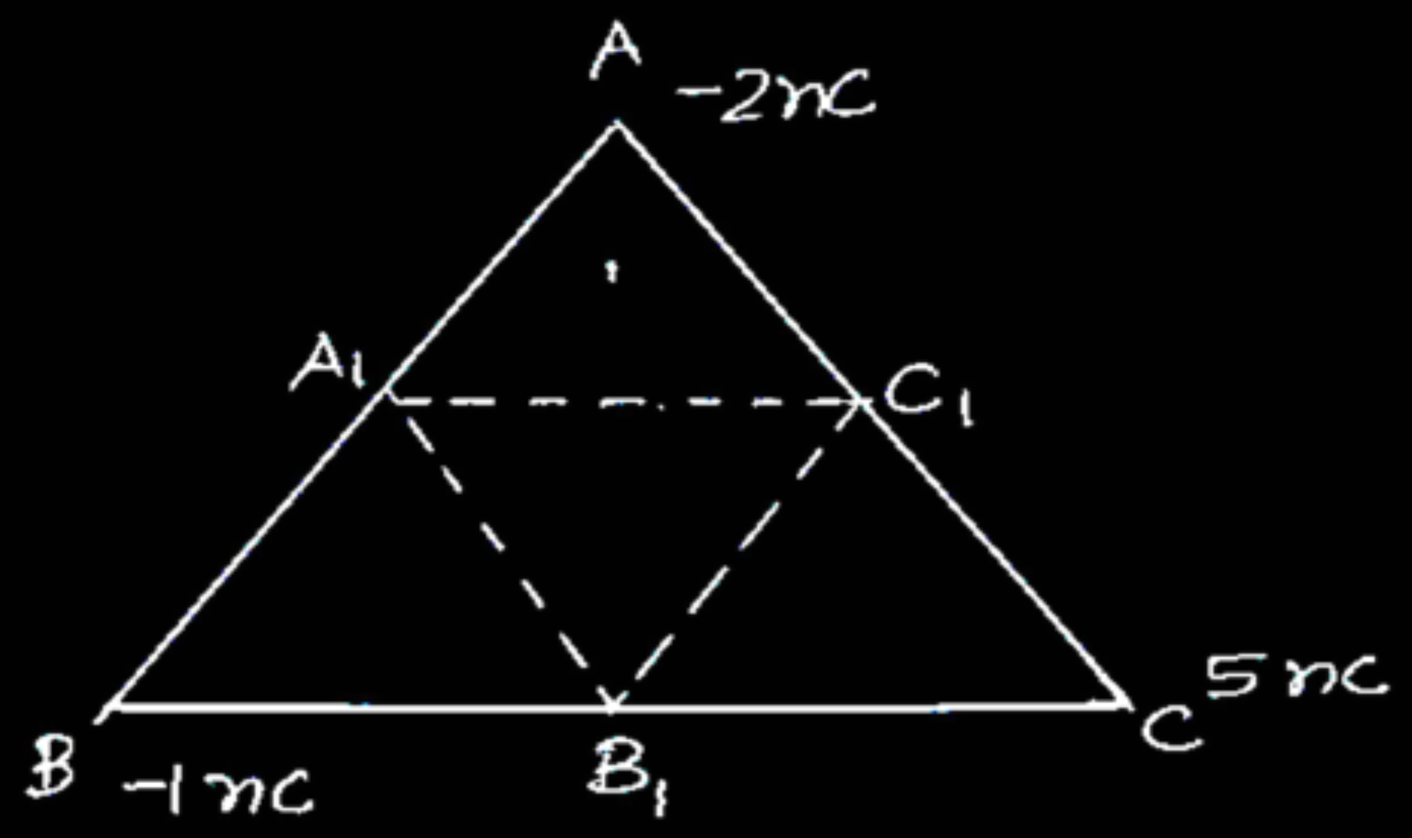
(A) $\frac{\alpha}{r}$

(ii) Three point charges of – 2 nC, – 1 nC, and + 5 nC are kept at the vertices A, B and C of an equilateral triangle of side 0·2 m. Find the total amount of work done in shifting the charges from A to A_1, B to B_1 and C to C_1. Here A_1, B_1 and C_1 are the midpoints of sides AB, BC and CA, respectively.

CBSE 25

(ii) Three point charges of – 2 nC, – 1 nC, and + 5 nC are kept at the vertices A, B and C of an equilateral triangle of side 0·2 m. Find the total amount of work done in shifting the charges from A to A_1, B to B_1 and C to C_1. Here A_1, B_1 and C_1 are the midpoints of sides AB, BC and CA, respectively.

CBSE 25

(ii) 	
Initial electrostatic potential energy of the system	
$U_1 = \frac{1}{4\pi\varepsilon_o}\left(\frac{q_A q_B}{AB} + \frac{q_C q_A}{AC} + \frac{q_C q_B}{BC}\right)$	
$= \frac{9\times10^9}{0.2}\left[(-2\times-1)+(-2\times5)+(-1\times5)\right]\times10^{-18}$	½
$U_1 = -5.85\times10^{-7}$ J	
$U_2 = \frac{1}{4\pi\varepsilon_o}\left(\frac{q_{A_1} q_{B_1}}{A_1B_1} + \frac{q_{C_1} q_{A_1}}{A_1C_1} + \frac{q_{c_1} q_{B_1}}{B_1C_1}\right)$	½
$U_2 = -11.7\times10^{-7}$ J	
$W = U_2 - U_1 = -5.85\times10^{-7}$ J	½

32. (a) (i) Two point charges 5 μC and –1 μC are placed at points (–3 cm, 0, 0) and (3 cm, 0, 0) respectively. An external electric field $\vec{E} = \frac{A}{r^2}\hat{r}$ where A = 3×10^5 Vm is switched on in the region. Calculate the change in electrostatic energy of the system due to the electric field. **5**

(ii) A system of two conductors is placed in air and they have net charge of +80μC and –80μC which causes a potential difference of 16 V between them.

(1) Find the capacitance of the system.

(2) If the air between the capacitor is replaced by a dielectric medium of dielectric constant 3, what will be the potential difference between the two conductors ?

(3) If the charges on two conductors are changed to +160 μC and –160 μC, will the capacitance of the system change ? Give reason for your answer.

CBSE 25

(i) $\vec{E} = \dfrac{3\times10^5}{r^2}\hat{r}$ (Given) $dV = -\vec{E}.\mathrm{d}\vec{r}$

$V = 3 \times 10^5 / r$

Electrostatic energy of the system in the absence of the field

$U_i = \dfrac{Kq_1q_2}{r_{12}}$ ½

Electrostatic energy in the presence of the field

$U_f = \dfrac{Kq_1q_2}{r_{12}} + q_1V(\vec{r}_1) + q_2V(\vec{r}_2)$

$\Delta U = U_f - U_i = q_1V(\vec{r}_1) + q_2V(\vec{r}_2)$ ½

$\Delta U = \dfrac{5\times10^{-6}\times3\times10^5}{3\times10^{-2}} - \dfrac{1\times10^{-6}\times3\times10^5}{3\times10^{-2}}$ ½

$= 40\ \mathrm{J}$ ½

ii) 1) $C = \dfrac{Q}{V} = \dfrac{80}{16} = 5\mu F$ 1

2) $C' = KC$

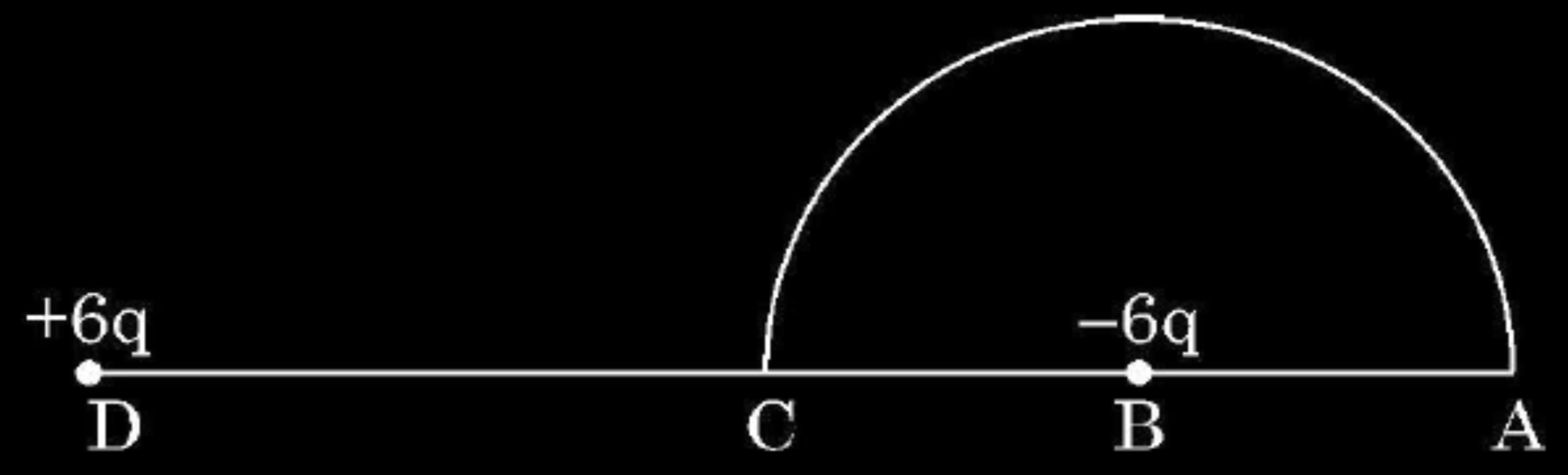
$= 3\times5\mu F = 15\ \mu F$ ½

$V' = \dfrac{Q}{C'} = \dfrac{80\mu C}{15\mu F} = 5.33V$ ½

3) No, ½

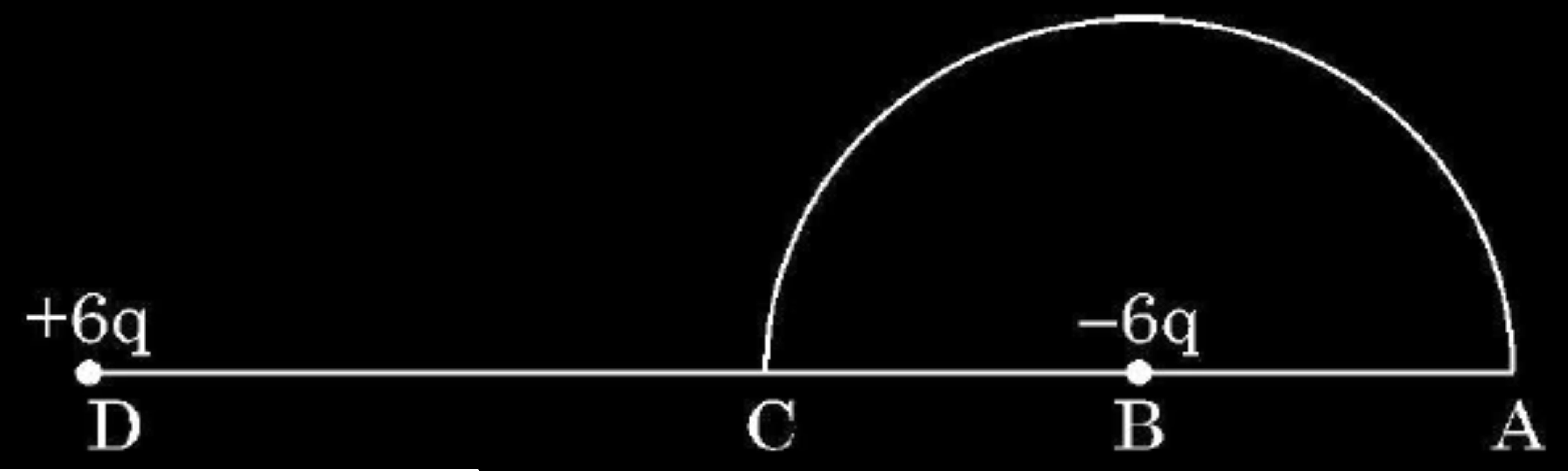
The capacitance of the system depends on its geometry. ½

(ii) A charge –6 μC is placed at the centre B of a semicircle of radius 5 cm, as shown in the figure. An equal and opposite charge is placed at point D at a distance of 10 cm from B. A charge +5 μC is moved from point 'C' to point 'A' along the circumference. Calculate the work done on the charge.



CBSE 25

(ii) A charge $-6\ \mu C$ is placed at the centre B of a semicircle of radius 5 cm, as shown in the figure. An equal and opposite charge is placed at point D at a distance of 10 cm from B. A charge $+5\ \mu C$ is moved from point 'C' to point 'A' along the circumference. Calculate the work done on the charge.



CBSE 25

ii) $V_c = \left[\dfrac{k\times 6\times 10^{-6}}{5\times 10^{-2}} - \dfrac{k\times 6\times 10^{-6}}{5\times 10^{-2}}\right]$	1
$= 0$	
$V_A = \left[\dfrac{k\times 6\times 10^{-6}}{15\times 10^{-2}} - \dfrac{k\times 6\times 10^{-6}}{5\times 10^{-2}}\right]$	½
$= \dfrac{k\times 6\times 10^{-6}}{10^{-2}}\left[\dfrac{1-3}{15}\right]$	
$= -\dfrac{9\times 10^{9}\times 6\times 10^{-6}\times 2}{15\times 10^{-2}}$	
$= -7.2\times 10^{5}$ V	½
$W = q\left[V_A - V_c\right]$	½
$= 5\times 10^{-6}\left[-7.2\times 10^{5} - 0\right]$	
$W = -3.6$ J	½

23. Two point charges of 10 μC and 20 μC are located at points (– 4 cm, 0, 0) and (5 cm, 0, 0) respectively, in a region with electric field $E = \frac{A}{r^2}$, where $A = 2 \times 10^6 \ NC^{-1} \ m^2$ and $\vec{r}$ is the position vector of the point under consideration. Calculate the electrostatic potential energy of the system. *3*

CBSE 24 C

23. Two point charges of 10 μC and 20 μC are located at points (– 4 cm, 0, 0) and (5 cm, 0, 0) respectively, in a region with electric field $E = \frac{A}{r^2}$, where $A = 2 \times 10^6\ NC^{-1}\ m^2$ and $\vec{r}$ is the position vector of the point under consideration. Calculate the electrostatic potential energy of the system. *3*

CBSE 24 C

Electrostatic potential energy of the system :	
$U = \frac{kq_1q_2}{r_{12}} + q_1V_1 + q_2V_2$	1
$\frac{kq_1q_2}{r_{12}} = \frac{9\times10^9 \times 10\times10^{-6} \times 20\times10^{-6}}{9\times10^{-2}} = 20J$	½
$q_1V_1 = q_1\frac{A}{r_1} = \frac{10\times10^{-6} \times 2\times10^6}{4\times10^{-2}} = 500J$	½
$q_2V_2 = q_2\frac{A}{r_2} = \frac{20\times10^{-6} \times 2\times10^6}{5\times10^{-2}} = 800J$	½
$U = (20+500+800)\,J$ $U = 1320\,J$	½

1. Two particles A and B of the same mass but having charges q and 4q respectively, are accelerated from rest through different potential differences V_A and V_B such that they attain same kinetic energies. The value of $\left(\frac{V_A}{V_B}\right)$ is :

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 4

CBSE 24 C

1. Two particles A and B of the same mass but having charges q and 4q respectively, are accelerated from rest through different potential differences V_A and V_B such that they attain same kinetic energies. The value of $\left(\frac{V_A}{V_B}\right)$ is :

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 4

CBSE 24 C

(D) 4

1. Three points charges + q, – 4q and + 2q are kept at the vertices of an equilateral triangle of side 'a'. The electrostatic potential energy of this system of charges is : CBSE 24 C

(A) $-\frac{1}{4\pi \varepsilon_0}\left(\frac{q^2}{a^2}\right)$

(B) $+\frac{1}{4\pi \varepsilon_0}\left(\frac{8q^2}{a^2}\right)$

(C) $+\frac{1}{4\pi \varepsilon_0}\left(\frac{6q^2}{a}\right)$

(D) $-\frac{1}{4\pi \varepsilon_0}\left(\frac{10q^2}{a}\right)$

1. Three points charges + q, – 4q and + 2q are kept at the vertices of an equilateral triangle of side 'a'. The electrostatic potential energy of this system of charges is :

CBSE 24 C

(A) $-\frac{1}{4\pi\varepsilon_0}\left(\frac{q^2}{a^2}\right)$

(B) $+\frac{1}{4\pi\varepsilon_0}\left(\frac{8q^2}{a^2}\right)$

(C) $+\frac{1}{4\pi\varepsilon_0}\left(\frac{6q^2}{a}\right)$

(D) $-\frac{1}{4\pi\varepsilon_0}\left(\frac{10q^2}{a}\right)$

(D) $-\frac{1}{4\pi\varepsilon_0}\left(\frac{10q^2}{a}\right)$

33. (a) (i) Draw equipotential surfaces for an electric dipole.

(ii) Two point charges q_1 and q_2 are located at $\vec{r}_1$ and $\vec{r}_2$ respectively in an external electric field $\vec{E}$. Obtain an expression for the potential energy of the system.

(iii) The dipole moment of a molecule is 10^{-30} Cm. It is placed in an electric field $\vec{E}$ of 10^5 V/m such that its axis is along the electric field. The direction of $\vec{E}$ is suddenly changed by $60°$ at an instant. Find the change in the potential energy of the dipole, at that instant.

CBSE 24

 1

(ii) Work done in bringing a charge q_1 from infinity to $\vec{r}_1$:

$$W_1 = q_1 V(\vec{r}_1) \qquad \text{-------------(1)}$$ ½

Work done in bringing a charge q_2 from infinity to $\vec{r}_2$ against the external field :

$$W_2 = q_2 V(\vec{r}_2) \qquad \text{-------------(2)}$$ ½

Work done on q_2 against the field due to q_1:

$$W_{12} = \frac{q_1 q_2}{4\pi\varepsilon_0 r_{12}} \qquad \text{-----------------(3)}$$ ½

Potential energy of the system = Total work done

$$= q_1 V(\vec{r}_1) + q_2 V(\vec{r}_2) + \frac{q_1 q_2}{4\pi\varepsilon_0 r_{12}}$$ ½

(iii) Change in Potential energy = Work done

$W = pE\,[\cos\theta_0 - \cos\theta_1]$ 1

$W = 10^{-30} \times 10^{5}\,[\cos 0^0 - \cos 60^0]$ ½

$W = 5.0 \times 10^{-26}$ J ½

1. An electric dipole of dipole moment $\vec{p}$ is kept in a uniform electric field $\vec{E}$. The amount of work done to rotate it from the position of stable equilibrium to that of unstable equilibrium will be **1**

(A) 2 pE (B) –2 pE

(C) pE (D) zero

CBSE 24

1. An electric dipole of dipole moment $\vec{p}$ is kept in a uniform electric field $\vec{E}$. The amount of work done to rotate it from the position of stable equilibrium to that of unstable equilibrium will be **1**

(A) 2 pE (B) –2 pE

(C) pE (D) zero

CBSE 24

(A) 2 pE

1. Three point charges, each of charge q are placed on vertices of a triangle ABC, with AB = AC = 5L, BC = 6L. The electrostatic potential at midpoint of side BC will be **1**

(A) $\frac{11}{48}\frac{q}{\pi\varepsilon_0 L}$

(B) $\frac{8q}{36\pi\varepsilon_0 L}$

(C) $\frac{5q}{24\pi\varepsilon_0 L}$

(D) $\frac{1}{16}\frac{q}{\pi\varepsilon_0 L}$

CBSE 24

1. Three point charges, each of charge q are placed on vertices of a triangle ABC, with AB = AC = 5L, BC = 6L. The electrostatic potential at midpoint of side BC will be **1**

(A) $\frac{11}{48}\frac{q}{\pi\varepsilon_0 L}$ (B) $\frac{8q}{36\pi\varepsilon_0 L}$

(C) $\frac{5q}{24\pi\varepsilon_0 L}$ (D) $\frac{1}{16}\frac{q}{\pi\varepsilon_0 L}$

CBSE 24

(A) $\frac{11}{48}\frac{q}{\pi\varepsilon_0 L}$

1. Two charges + q each are kept '2a' distance apart. A third charge – 2q is placed midway between them. The potential energy of the system is – **1**

(A) $\frac{q^2}{8\pi\varepsilon_0 a}$ (B) $-\frac{6q^2}{8\pi\varepsilon_0 a}$

(C) $\frac{-7q^2}{8\pi\varepsilon_0 a}$ (D) $\frac{9q^2}{8\pi\varepsilon_0 a}$

CBSE 24

1. Two charges + q each are kept '2a' distance apart. A third charge – 2q is placed midway between them. The potential energy of the system is – **1**

(A) $\frac{q^2}{8\pi\varepsilon_0 a}$ (B) $-\frac{6q^2}{8\pi\varepsilon_0 a}$

(C) $\frac{-7q^2}{8\pi\varepsilon_0 a}$ (D) $\frac{9q^2}{8\pi\varepsilon_0 a}$

CBSE 24

(C) $\frac{-7q^2}{8\pi\varepsilon_0 a}$

2. A proton is taken from point P_1 to point P_2, both located in an electric field. The potentials at points P_1 and P_2 are -5 V and $+5$ V respectively. Assuming that kinetic energies of the proton at points P_1 and P_2 are zero, the work done on the proton is :

(A) $-1{\cdot}6 \times 10^{-18}$ J

(B) $1{\cdot}6 \times 10^{-18}$ J

(C) Zero

(D) $0{\cdot}8 \times 10^{-18}$ J

CBSE 24

2. A proton is taken from point P_1 to point P_2, both located in an electric field. The potentials at points P_1 and P_2 are -5 V and $+5$ V respectively. Assuming that kinetic energies of the proton at points P_1 and P_2 are zero, the work done on the proton is :

(A) $-1{\cdot}6 \times 10^{-18}$ J

(B) $1{\cdot}6 \times 10^{-18}$ J

(C) Zero

(D) $0{\cdot}8 \times 10^{-18}$ J

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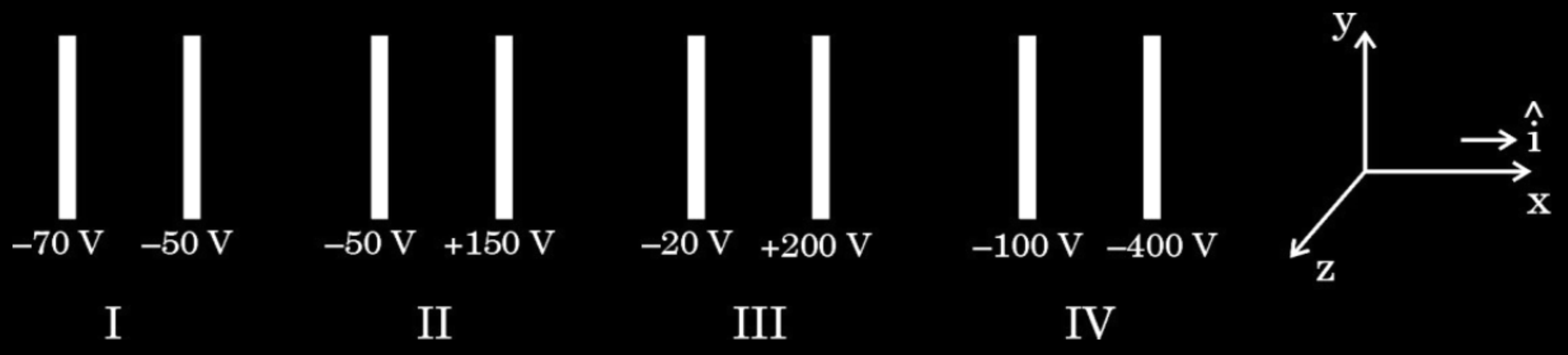
(B) 1.6×10^{-18} J

29. The figure shows four pairs of parallel identical conducting plates, separated by the same distance 2·0 cm and arranged perpendicular to x-axis. The electric potential of each plate is mentioned. The electric field between a pair of plates is uniform and normal to the plates. CBSE 24

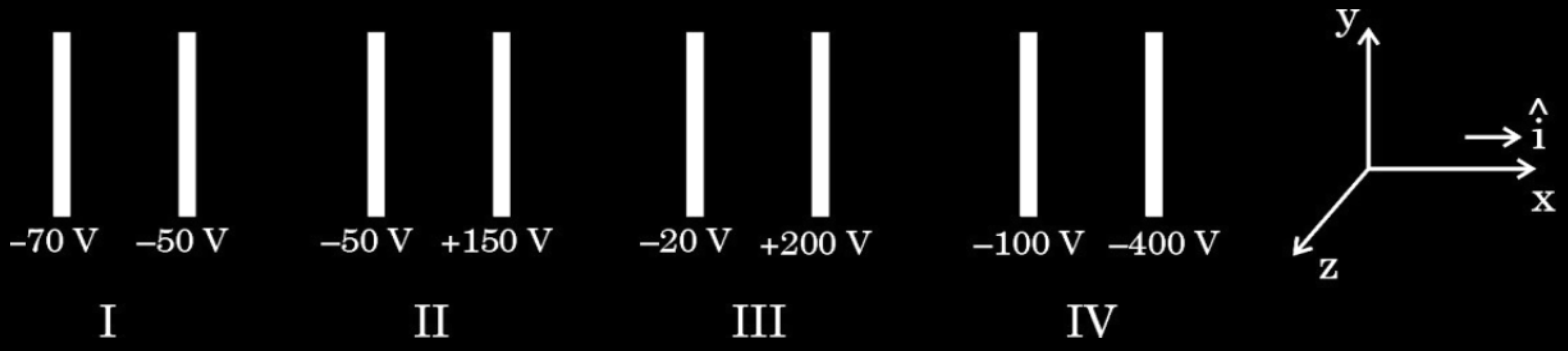


(i) For which pair of the plates is the electric field $\vec{E}$ along $\hat{i}$? *1*

(A) I (B) II

(C) III (D) IV

(D) IV



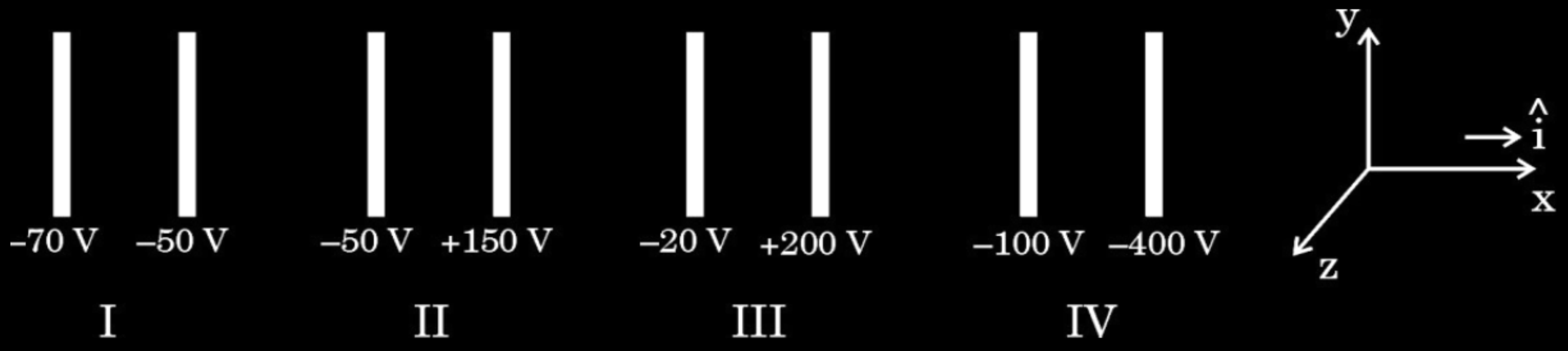
(ii) An electron is released midway between the plates of pair IV. It will : 1

(A) move along $\hat{i}$ at constant speed

(B) move along $-\hat{i}$ at constant speed

(C) accelerate along $\hat{i}$

(D) accelerate along $-\hat{i}$



(ii) An electron is released midway between the plates of pair IV. It will : *1*

(A) move along $\hat{i}$ at constant speed

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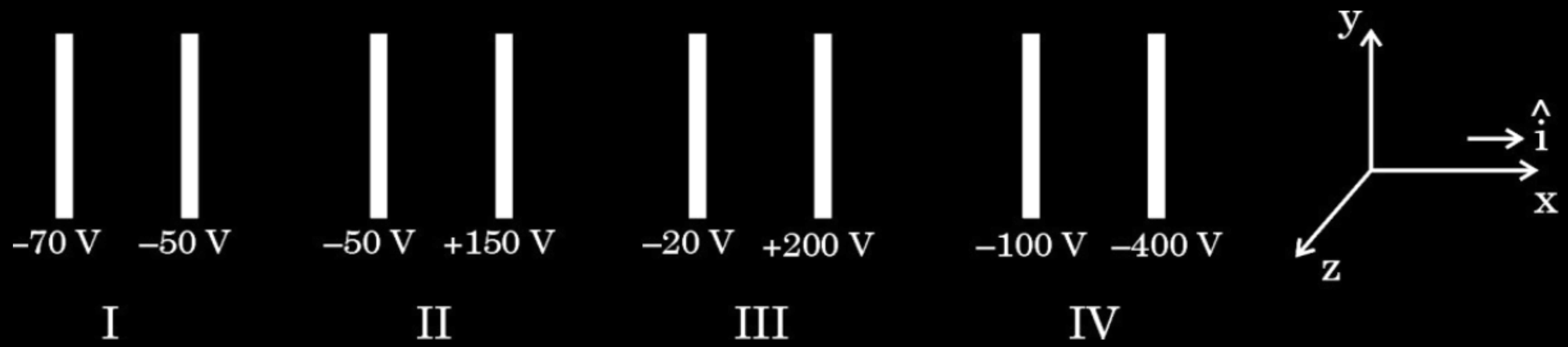
(C) accelerate along $\hat{i}$

(D) accelerate along $-\hat{i}$

(D) accelerate along $-\hat{i}$



(iii) Let V_0 be the potential at the left plate of any set, taken to be at $x = 0$ m. Then potential V at any point $(0 \le x \le 2$ cm) between the plates of that set can be expressed as : *1*

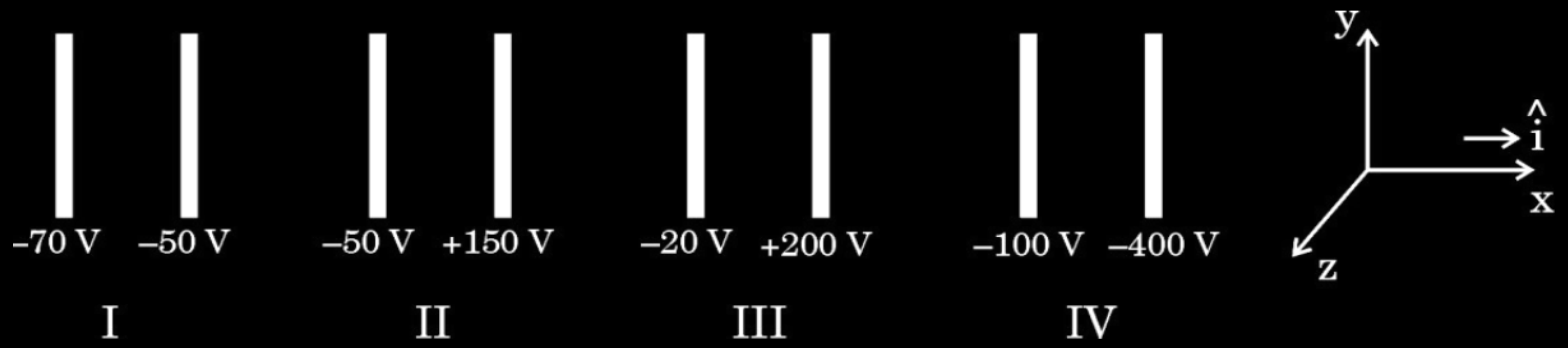
(A) $V = V_0 + \alpha x$

(B) $V = V_0 + \alpha x^2$

(C) $V = V_0 + \alpha x^{1/2}$

(D) $V = V_0 + \alpha x^{3/2}$

where α is a constant, positive or negative.



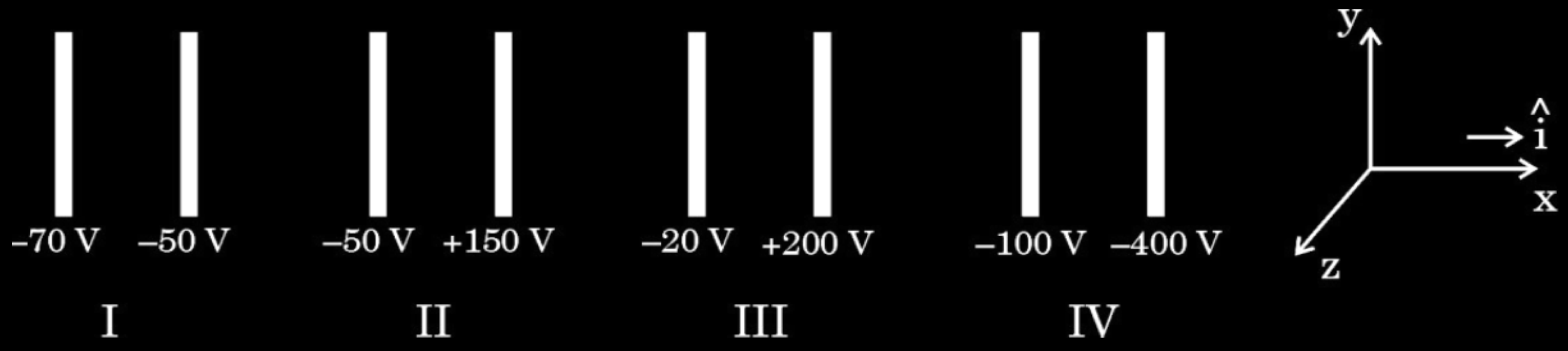
(iii) Let V_0 be the potential at the left plate of any set, taken to be at $x = 0$ m. Then potential V at any point ($0 \le x \le 2$ cm) between the plates of that set can be expressed as : *1*

(A) $V = V_0 + \alpha x$ (B) $V = V_0 + \alpha x^2$

(C) $V = V_0 + \alpha x^{1/2}$ (D) $V = V_0 + \alpha x^{3/2}$

where α is a constant, positive or negative.

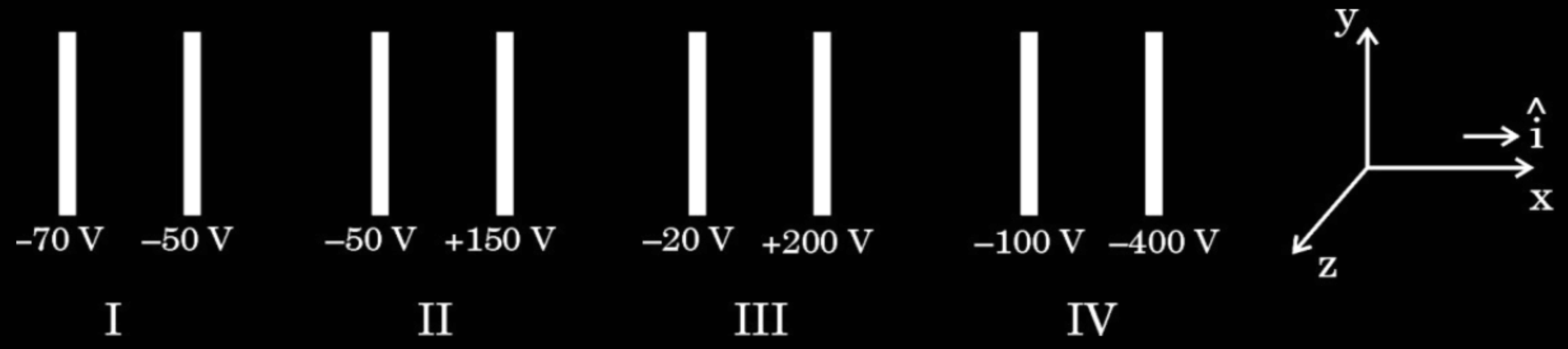
(iii) (A) $V = V_0 + \alpha x$



(iv) (a) Let E_1, E_2, E_3 and E_4 be the magnitudes of the electric field between the pairs of plates, I, II, III and IV respectively. Then : *1*

(A) $E_1 > E_2 > E_3 > E_4$ (B) $E_3 > E_4 > E_1 > E_2$

(C) $E_4 > E_3 > E_2 > E_1$ (D) $E_2 > E_3 > E_4 > E_1$

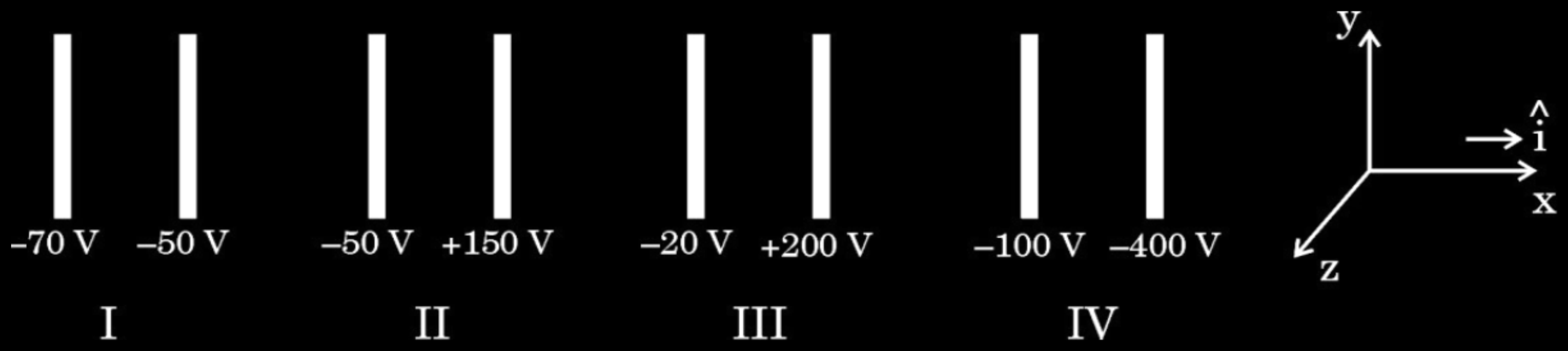


(iv) (a) Let E_1, E_2, E_3 and E_4 be the magnitudes of the electric field between the pairs of plates, I, II, III and IV respectively. Then : *1*

(A) $E_1 > E_2 > E_3 > E_4$ (B) $E_3 > E_4 > E_1 > E_2$

(C) $E_4 > E_3 > E_2 > E_1$ (D) $E_2 > E_3 > E_4 > E_1$

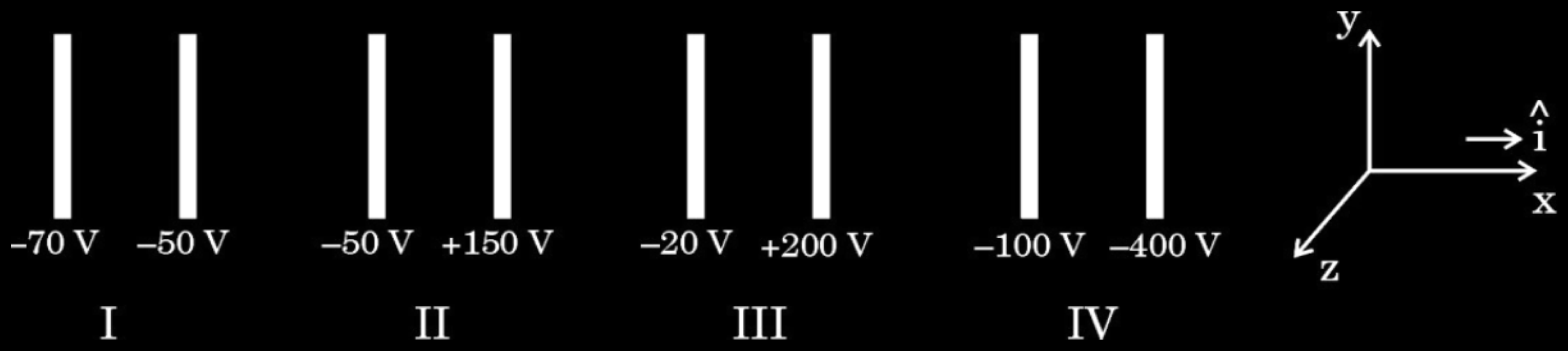
(a) (C) $E_4 > E_3 > E_2 > E_1$



(b) An electron is projected from the right plate of set I directly towards its left plate. It just comes to rest at the plate. The speed with which it was projected is about :
(Take (e/m) $= 1\cdot76 \times 10^{11}$ C/kg) *1*

(A) $1\cdot3 \times 10^{5}$ m/s (B) $2\cdot6 \times 10^{6}$ m/s

(C) $6\cdot5 \times 10^{5}$ m/s (D) $5\cdot2 \times 10^{7}$ m/s



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(Take (e/m) $=1.76 \times 10^{11}$ C/kg) *1*

(A) 1.3×10^5 m/s (B) 2.6×10^6 m/s

(C) 6.5×10^5 m/s (D) 5.2×10^7 m/s

(B) $2.6 \text{x} 10^6$ m/s

(ii) Three point charges q, 2q and nq are placed at the vertices of an equilateral triangle. If the potential energy of the system is zero, find the value of n.

CBSE 24

$$\frac{1}{r_1} \cong \frac{1}{r}\left(1-\frac{2a\cos\theta}{r}\right)^{-1/2}$$

$$\cong \frac{1}{r}\left(1+\frac{a\cos\theta}{r}\right) \quad \text{------------------------------------(2)}$$

$$\frac{1}{r_2} \cong \frac{1}{r}\left(\frac{1+2a\cos\theta}{r}\right)^{-1/2} \quad \text{------------------------------(3)}$$

$$\cong \frac{1}{r}\left(1-\frac{a\cos\theta}{r}\right)$$ ½

Using eqn. (1) (2), (3) and p = 2qa

$$V = \frac{q}{4\pi\varepsilon_0}\frac{2a\cos\theta}{r^2}$$ ½

$$= \frac{\mathrm{p}\cos\theta}{4\pi\varepsilon_0 r^2}$$

(ii) Consider the side of equilateral triangle as 'a' ½

Potential energy $= \text{U} = \dfrac{kq_1q_2}{a}+\dfrac{kq_2q_3}{a}+\dfrac{kq_1q_3}{a}$ ½

According to question

$$U = \frac{k(q)(2q)}{a}+\frac{k(2q)(nq)}{a}+\frac{k(q)(\mathrm{n}\,q)}{a} = 0$$ ½

$$= \frac{2q^2}{a}+\frac{2nq^2}{a}+\frac{\mathrm{n}\,q^2}{a} = 0$$

$$2+2n+n=0$$ ½

$$3n=-2$$

$$n=-\frac{2}{3}$$ ½

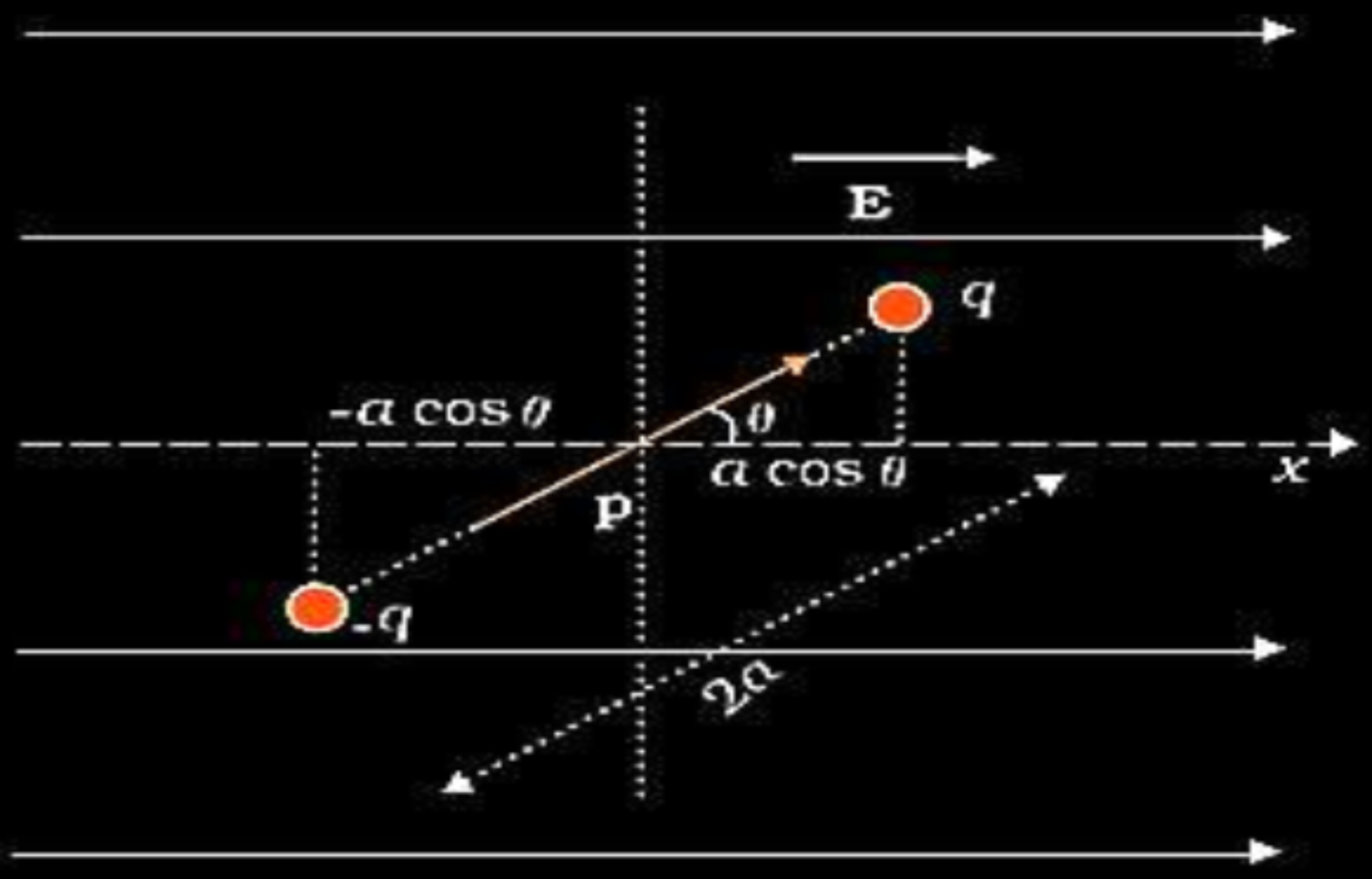
32. (a) (i) Derive an expression for potential energy of an electric dipole $\vec{p}$ in an external uniform electric field $\vec{E}$. When is the potential energy of the dipole (1) maximum, and (2) minimum ?

CBSE 24

32. (a) (i) Derive an expression for potential energy of an electric dipole $\vec{p}$ in an external uniform electric field $\vec{E}$. When is the potential energy of the dipole (1) maximum, and (2) minimum ?

CBSE 24

The amount of work done in rotating the dipole from $\theta = \theta_0$ to $\theta = \theta_1$ by the external torque



$$W = \int_{\theta_0}^{\theta_1} \tau_{ext}\, d\theta$$ ½

$$= \int_{\theta_0}^{\theta_1} pE\sin\theta\ d\theta$$ ½

$$W = pE(\cos\theta_0 - \cos\theta_1)$$

For $\theta_0 = \dfrac{\pi}{2}$ & $\theta_1 = \theta$ ½

$$= pE(\cos\frac{\pi}{2} - \cos\theta)$$

$$U(\theta) = -pE\ \cos\theta$$

$$= -\vec{p}.\vec{E}$$ ½

(1) Potential energy is maximum when:
$\vec{p}$ is antiparallel to $\vec{E}$ ½
Alternatively:
$\theta = 180°$ or π radians

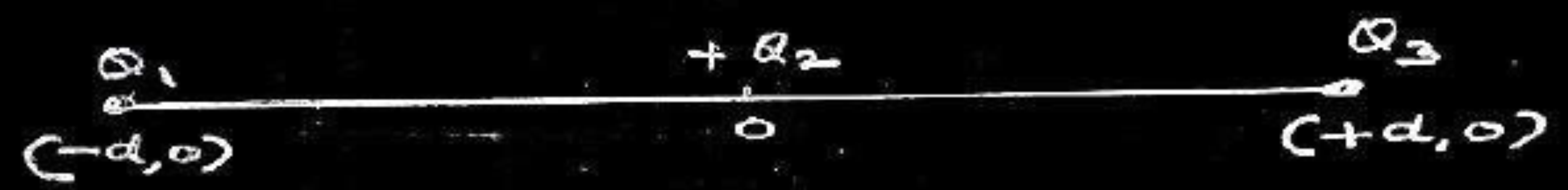
(2) Potential energy is minimum when:
$\vec{p}$ is along to $\vec{E}$ ½
Alternatively:
$\theta = 0°$

22. Three point charges Q_1, Q_2 and Q_3 are located in $x - y$ plane at points $(-d, 0)$, $(0, 0)$ and $(d, 0)$ respectively. Q_1 and Q_3 are identical and Q_2 is positive. What will be the nature and value of Q_1 so that the potential energy of the system is zero ? *3*

CBSE 24

22. Three point charges Q_1, Q_2 and Q_3 are located in x – y plane at points (– d, 0), (0, 0) and (d, 0) respectively. Q_1 and Q_3 are identical and Q_2 is positive. What will be the nature and value of Q_1 so that the potential energy of the system is zero ? *3*

CBSE 24



Nature of Q_1 will be negative. 1

Let , $Q_1 = Q_3 = q$ ½

$$\frac{1}{4\pi\varepsilon_0}\left[\frac{qQ_2}{d}+\frac{qq}{2d}+\frac{Q_2 q}{d}\right]=0$$

$$\frac{1}{4\pi\varepsilon_0 d}\left[qQ_2+\frac{q^2}{2}+Q_2 q\right]=0$$

½

$$2qQ_2+\frac{q^2}{2}=0$$

$$2qQ_2=-\frac{q^2}{2}$$

$$Q_1=q=-4Q_2$$ 1

(b) (i) Obtain an expression for electric potential at a distance r from a point charge Q.

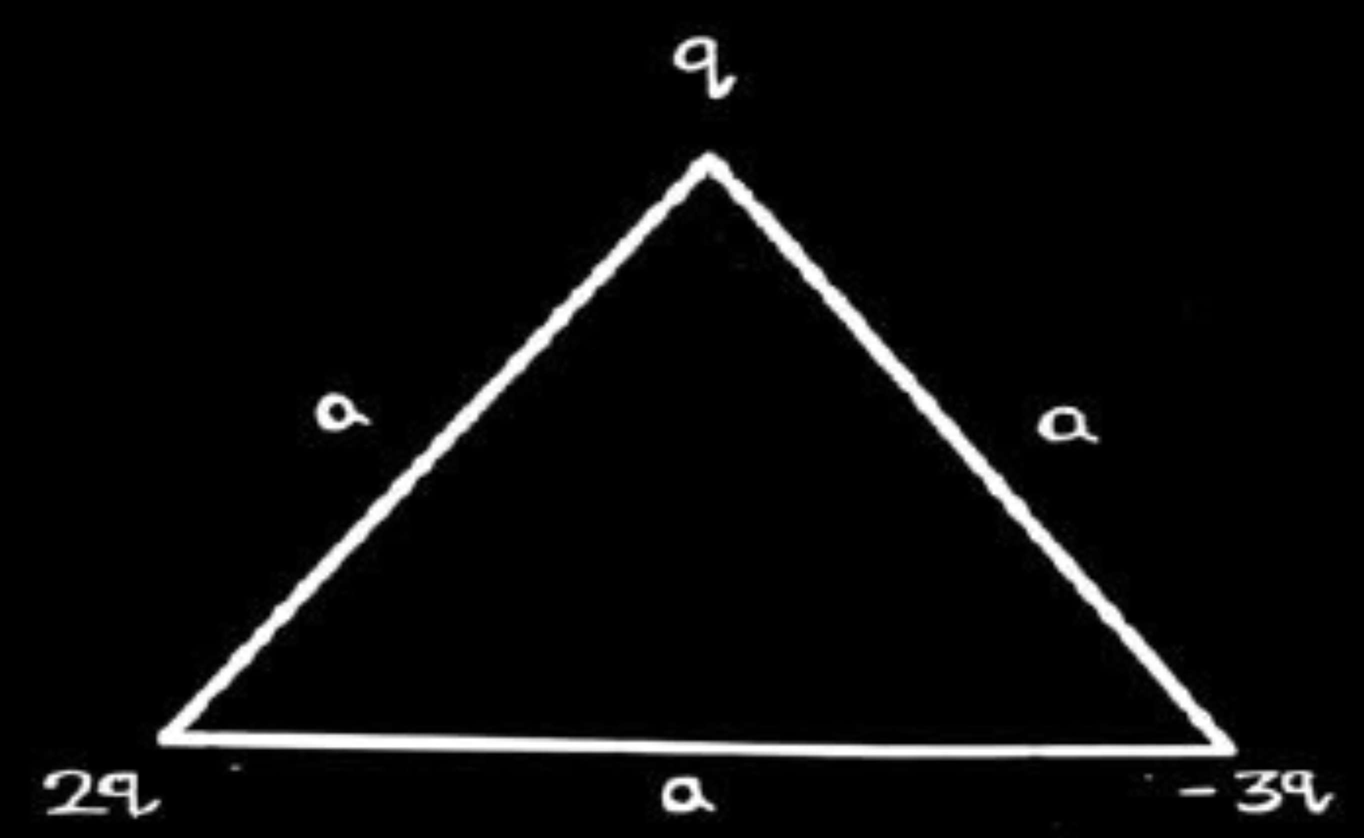
(ii) Two point charges of $10\ \mu C$ and $-5\ \mu C$ are placed at $(-3\ \text{cm}, 0)$ and $(6\ \text{cm}, 0)$ in an external electric field $E = \dfrac{A}{r^2}$, where $A = 1{\cdot}8 \times 10^5\ \text{N C}^{-1}\ \text{m}^2$. Calculate the electrostatic energy of the system. CBSE 24 *5*

(i) At some intermediate point P' the electrostatic force on unit positive test charge $= \frac{Q}{4\pi\varepsilon_0 r'^2}$	½
Work done in bringing a unit charge against this force is $\Delta W = -\frac{Q}{4\pi\varepsilon_0 r'^2}\Delta r'$	½
Total work done (W) by the external force is $W = -\int_{\infty}^{r}\frac{Q}{4\pi\varepsilon_0 r'^2}dr' = \frac{Q}{4\pi\varepsilon_0 r}$	1
This work done is the potential at point P due to charge Q. $V(r) = \frac{Q}{4\pi\varepsilon_0 r}$	1

(ii) $V = -\int \vec{E}.d\vec{r}$	½
$U_1 = q_1V_1 = \frac{A\times10\times10^{-6}}{3\times10^{-2}}$ $= \frac{1.8\times10^5\times10\times10^{-6}}{3\times10^{-2}}$ $= 60J$	
$U_2 = q_2V_2 = \frac{A\times(-5)\times10^{-6}}{6\times10^{-2}}$ $= \frac{1.8\times10^5\times(-5)\times10^{-6}}{6\times10^{-2}}$ $U_2 = -15J$	½
$U_{12} = \frac{Kq_1q_2}{r_{12}} = \frac{9\times10^9\times10\times10^{-6}\times(-5\times10^{-6})}{9\times10^{-2}} = -5J$	½
Total Energy $= U_1+U_2+U_3 = 60-15-5 = 40J$	½

24. (a) Obtain an expression for electrostatic potential energy of a system of three charges q, 2q and –3q placed at the vertices of an equilateral triangle of side a. CBSE 23 *2*

24. (a) Obtain an expression for electrostatic potential energy of a system of three charges q, 2q and –3q placed at the vertices of an equilateral triangle of side a. CBSE 23 *2*

 **OR a similar diagram with different order of charges**	
$U = \dfrac{1}{4\pi \in_0} \cdot \dfrac{q_1 q_2}{r}$	½
$U = \dfrac{1}{4\pi \in_0}\left[\dfrac{2q^2}{a} - \dfrac{6q^2}{a} - \dfrac{3q^2}{a}\right]$	1
$U = \dfrac{1}{4\pi \in_0} \dfrac{\left(-7q^2\right)}{a}$	½

(c) An electric dipole of dipole moment of 6×10^{-7} C-m is kept in a uniform electric field of 10^4 N/C such that the dipole moment and the electric field are parallel. Calculate the potential energy of the dipole. *2*

OR

(c) An electric dipole of dipole moment $\vec{p}$ is initially kept in a uniform electric field $\vec{E}$ such that $\vec{p}$ is perpendicular to $\vec{E}$. Find the amount of work done in rotating the dipole to a position at which $\vec{p}$ becomes antiparallel to $\vec{E}$. CBSE 23 *2*

(c) $U = -pE\cos\theta$ ½

$\theta = 0^0$ ½

$U = -(6 \times 10^{-7}) \times (10^4)$ 1

$U = -6 \times 10^{-3}$ J

OR

$\because$ Work done $W = -pE(\cos\theta_2 - \cos\theta_1)$ ½

where $\theta_2 = 180^\circ$, $\theta_1 = 90^\circ$ ½

$\Rightarrow W = -pE(\cos 180^\circ - \cos 90^\circ)$ ½

$W = +pE$ ½

(iii) Two point charges + q and – q are located at points (3a, 0) and (0, 4a) respectively in x-y plane. A third charge Q is kept at the origin. Find the value of Q, in terms of q and a, so that the electrostatic potential energy of the system is zero. *5*

CBSE 23

(iii)

Potential energy of the system = $K\left[\frac{Q(-q)}{4a}+\frac{Qq}{3a}-\frac{q^2}{5a}\right]$ ½

Potential energy of the system = 0 ½

$$\Rightarrow K\left[\frac{-Qq}{4a}+\frac{Qq}{3a}-\frac{q^2}{5a}\right]=0$$

$$\Rightarrow \frac{-Q}{4}+\frac{Q}{3}-\frac{q}{5}=0$$

$$\Rightarrow +\frac{Q}{12}-\frac{q}{5}=0$$ 1

$$\Rightarrow Q=+\ \frac{12q}{5}$$

34. A beam of electrons moving horizontally with a velocity of 3×10^7 m/s enters a region between two plates as shown in the figure. A suitable potential difference is applied across the plates such that the electron beam just strikes the edge of the lower plate.

CBSE 23

Answer the following questions based on the above :

(a) How long does an electron take to strike the edge ? *1*

(b) What is the shape of the path followed by the electron and why ? *1*

(c) Find the potential difference applied. *2*

OR

(c) Find the magnitude and direction of the magnetic field which should be created in the space between the plates so that the electron beam goes straight undeviated. *2*

(a) Electron strikes the edge after travelling 3 cm horizontally (along x-axis).	
$S_x = v_x \times t$	½
$3 \times 10^{-2} = (3 \times 10^7) \times t$	
$t = 10^{-9}$ s	½
(b) Shape of the path is parabola.	½
Reason: Force/acceleration is in a fixed direction perpendicular to the initial velocity.	½
(c) Along y-direction	
$S_y = u_y t + \frac{1}{2} a_y t^2$	
$-0.5 \times 10^{-2} = 0 + \frac{1}{2} a_y (10^{-9})^2$	
$a_y = -10^{16}$ m/s^2	½
Magnitude of acceleration $(a_y) = \frac{eE}{m} = \frac{e}{m}\left(\frac{V}{l}\right)$	½
$V = \frac{10^{16} \times 9.1 \times 10^{-31} \times 1 \times 10^{-2}}{1.6 \times 10^{-19}}$	½
$V = 568.75$ V	½

(c) $\lvert qE \rvert = \lvert qvB \rvert$; $B = \frac{E}{v} = (\frac{ma_y}{e})(\frac{1}{v})$	½
Along y-direction	
$S_y = u_y t + \frac{1}{2} a_y t^2$	
$-0.5 \times 10^{-2} = 0 + \frac{1}{2} a_y (10^{-9})^2$	
$a_y = -10^{16}$ m/s^2	½
$B = \left(\frac{9.1 \times 10^{-31} \times 10^{16}}{1.6 \times 10^{-19}}\right) \times \left(\frac{1}{3 \times 10^7}\right)$	½
$B = 1.9 \times 10^{-3} T$	½
Direction of magnetic field will be out of the plane of the paper.	

28. Three point charges Q_1 ($-15\ \mu C$), Q_2 ($10\ \mu C$) and Q_3 ($16\ \mu C$) are located at (0 cm, 0 cm), (0 cm, 3 cm) and (4 cm, 3 cm) respectively. Calculate the electrostatic potential energy of this system of charges. $\boxed{\text{CBSE 23}}$ *3*

28. Three point charges Q_1 (–15 μC), Q_2 (10 μC) and Q_3 (16 μC) are located at (0 cm, 0 cm), (0 cm, 3 cm) and (4 cm, 3 cm) respectively. Calculate the electrostatic potential energy of this system of charges. CBSE 23 *3*

$U_{Q1Q2} = \frac{1}{4\pi\epsilon_o}\frac{Q_1Q_2}{r_{12}}$	½
$= \frac{9\times10^9\times(-15\times10^{-6})\times(10\times10^{-6})}{3\times10^{-2}}$	½
$= -45$ J	
$U_{Q2Q3} = \frac{1}{4\pi\epsilon_o}\frac{Q_2Q_3}{r_{23}}$	
$= \frac{9\times10^9\times(10\times10^{-6})\times(16\times10^{-6})}{4\times10^{-2}}$	½
$= 36$ J	
$U_{Q1Q3} = \frac{1}{4\pi\epsilon_o}\frac{Q_1Q_3}{r_{13}}$	
$= \frac{9\times10^9\times(-15\times10^{-6})\times(16\times10^{-6})}{5\times10^{-2}}$	½
$= -43.2$ J	½
$U_{net} = U_{Q1Q2} + U_{Q2Q3} + U_{Q1Q3}$	½
$= -45+36-43.2$	
$= -52.2$ J	½
Alternatively	
$U = k(\frac{Q_1Q_2}{r_{12}} + \frac{Q_2Q_3}{r_{23}} + \frac{Q_1Q_3}{r_{13}})$	1
$= \frac{9\times10^9\times(-15\times10^{-6})\times(10\times10^{-6})}{3\times10^{-2}} + \frac{9\times10^9\times(10\times10^{-6})\times(16\times10^{-6})}{4\times10^{-2}} + \frac{9\times10^9\times(-15\times10^{-6})\times(16\times10^{-6})}{5\times10^{-2}}$	1
$=(-45 +36-43.2)$ J	
$= 52.2$ J	1

(b) (i) Consider two identical point charges located at points (0, 0) and (a, 0).

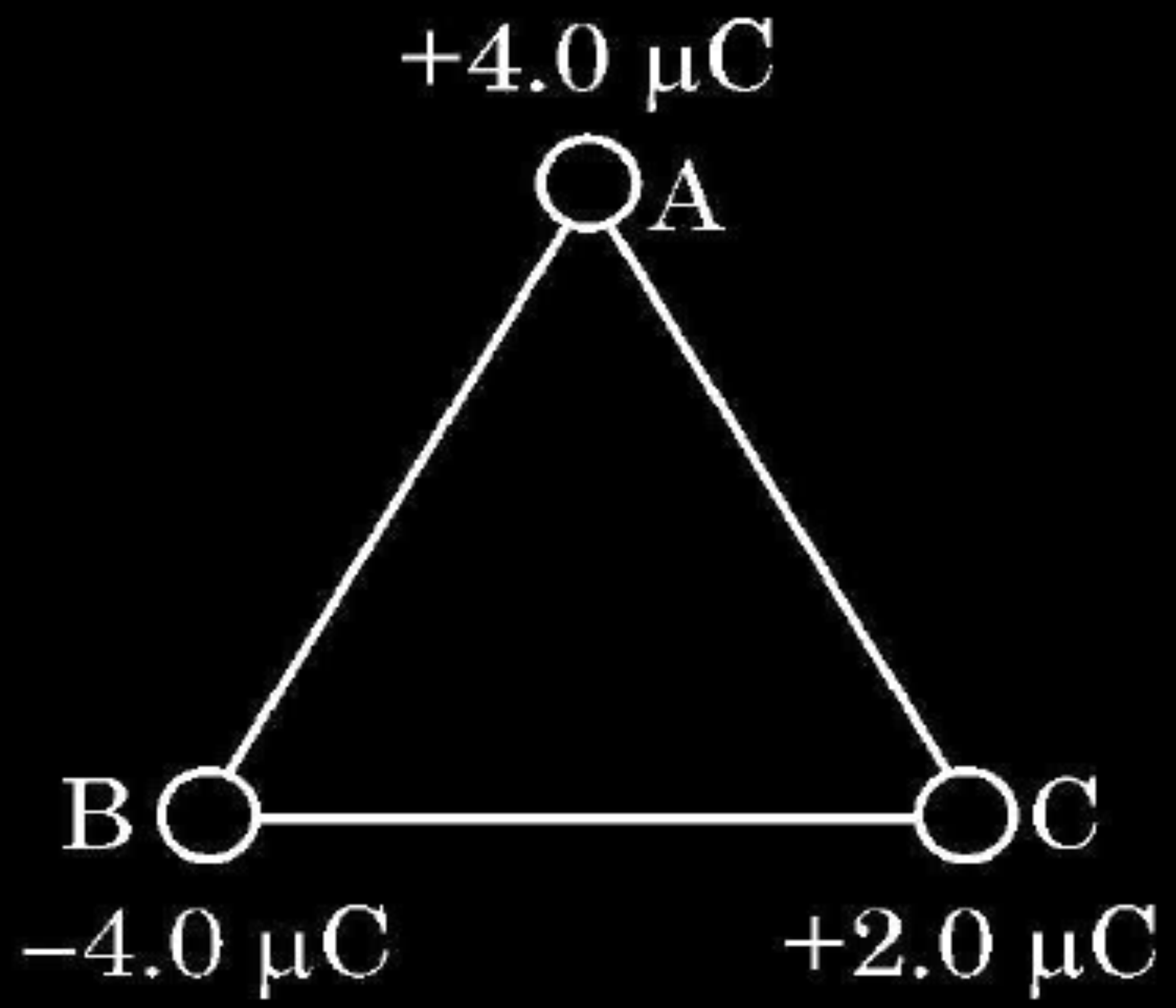
(1) Is there a point on the line joining them at which the electric field is zero ?

(2) Is there a point on the line joining them at which the electric potential is zero ?

Justify your answers for each case.

(ii) State the significance of negative value of electrostatic potential energy of a system of charges.

Three charges are placed at the corners of an equilateral triangle ABC of side 2.0 m as shown in figure. Calculate the electric potential energy of the system of three charges.



CBSE 23

(i) (1) Yes , electric field is zero at mid point.

Electric field being a vector quantity , its resultant is zero. ½

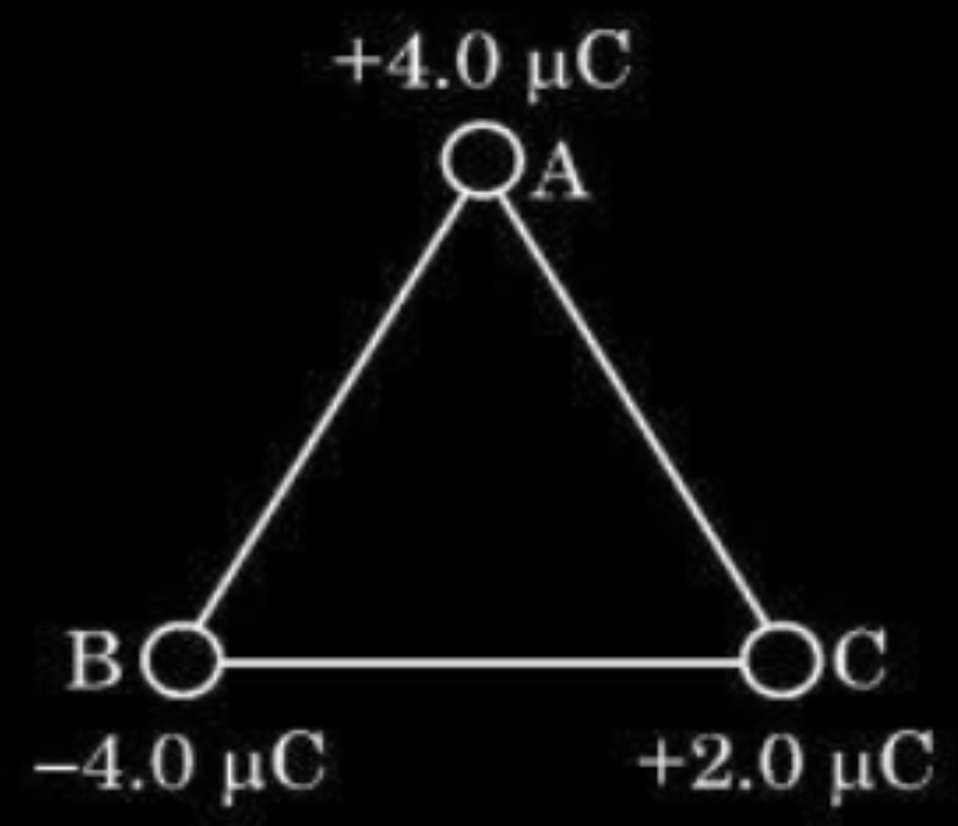
(2) No, potential cannot be zero on line joining the charges. ½

Electric potential being a scalar quantity, the net potential due to two identical charges cannot be zero. ½

(ii) Negative value of electrostatic potential energy of a system signifies that the system has attractive forces. ½

Alternatively 1

Give full credit , if a candidate writes the system is stable /bound.



$$U = \frac{1}{4\pi \in_0} \times \frac{q_1 q_2}{r}$$

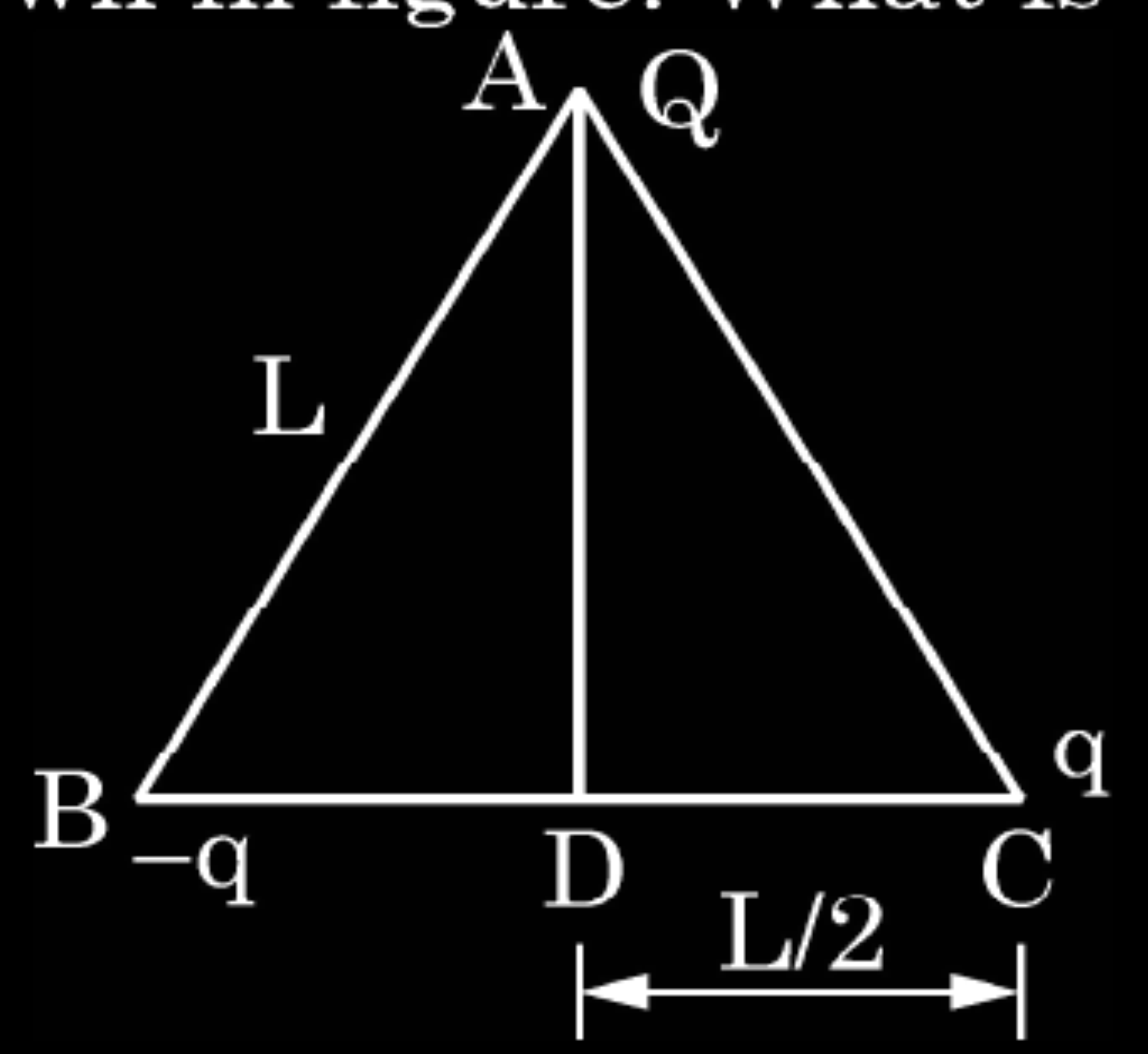
$$U = \frac{1}{4\pi \in_0}\left[\frac{q_A q_B}{r} + \frac{q_B q_c}{r} + \frac{q_c q_A}{r}\right]$$ ½

½

$$= \frac{9\times10^{9}}{2}\left[-16-8+8\right]\times10^{-12}$$ ½

$$= -7.2\times10^{-2}\,J$$ ½

22. Three point charges Q, q and – q are kept at the vertices of an equilateral triangle of side L as shown in figure. What is **2**

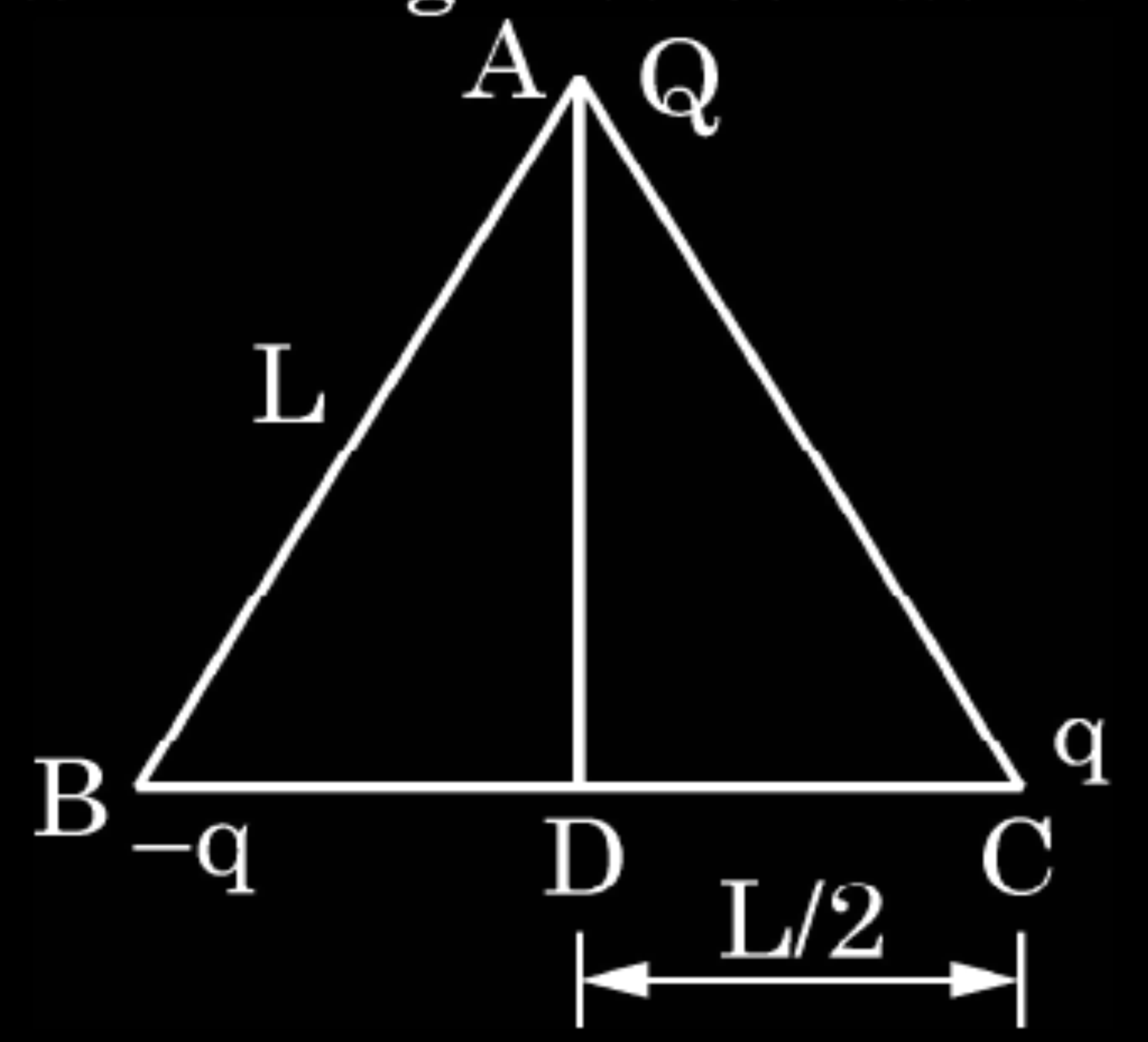


CBSE 23

(i) the electrostatic potential energy of the arrangement ? and

(ii) the potential at point D ?

22. Three point charges Q, q and – q are kept at the vertices of an equilateral triangle of side L as shown in figure. What is **2**



CBSE 23

(i) the electrostatic potential energy of the arrangement ? and

(ii) the potential at point D ?

(i) $U = -\frac{kQq}{L} + \frac{kQq}{L} - \frac{kq^2}{L}$	½
$U = -\frac{kq^2}{L}$	½
(ii) $V = V_{DA} + V_{DB} + V_{DC}$	
$V = \frac{2KQ}{L\sqrt{3}} - \frac{2Kq}{L} + \frac{2Kq}{L}$	½
$V = \frac{2KQ}{L\sqrt{3}}$	½

33. (a) Two point charges q_1 and q_2 are kept at a distance of r_{12} in air. Deduce the expression for the electrostatic potential energy of this system.

(b) If an external electric field (E) is applied on the system, write the expression for the total energy of this system. 3

CBSE 20

33. (a) Two point charges q_1 and q_2 are kept at a distance of r_{12} in air. Deduce the expression for the electrostatic potential energy of this system.

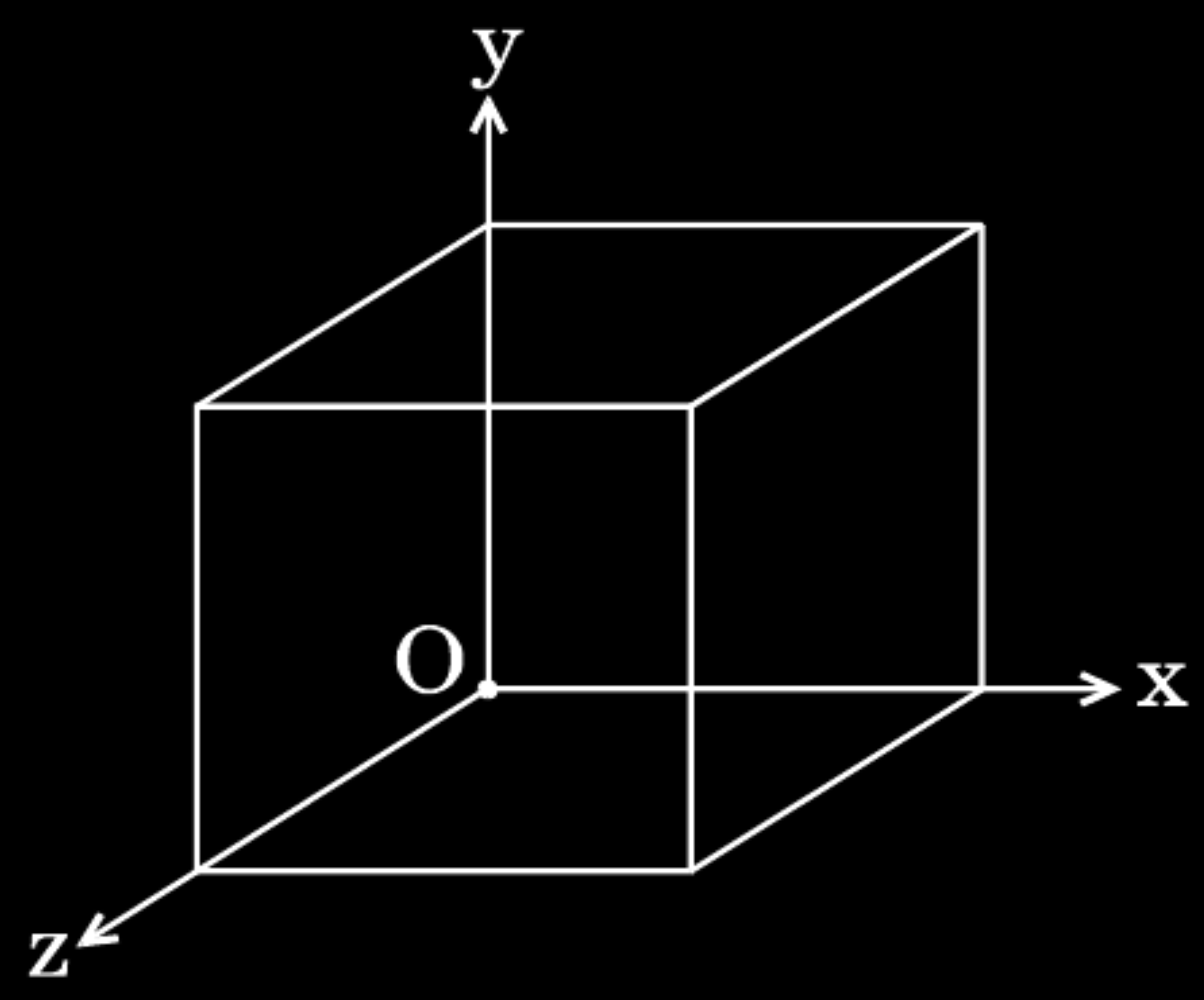
(b) If an external electric field (E) is applied on the system, write the expression for the total energy of this system. **3**

CBSE 20

(a) Work done in bringing the charge q_2, from infinity, to a point $= q_2 \times$ potential at the point due to charge q_1 $= q_2 \times \dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1}{r_{12}}$	½
$\therefore$ potential energy of the system $= \dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1 q_2}{r_{12}}$	1
(b)Let the potentials, at two points, due to an external electric field (E) be V_1 and V_2 respectively. Now the total energy of the system is: $\left[q_1V_1 + q_2V_2 + \dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1q_2}{r_{12}}\right]$	1½

(a) Two point charges q_1 and q_2 are kept r distance apart in a uniform external electric field $\vec{E}$. Find the amount of work done in assembling this system of charges.

(b) A cube of side 20 cm is kept in a region as shown in the figure. An electric field $\vec{E}$ exists in the region such that the potential at a point is given by $V = 10x + 5$, where V is in volt and x is in m.



CBSE 20

Find the

(i) electric field $\vec{E}$, and

(ii) total electric flux through the cube. 5

The work done in bringing charge q_1 from infinity to r_1 is	
$$W_1 = q_1V_1$$	½
The work done in bringing charge q_2 from infinity to r_2 is	
$$W_2 = q_2V_2$$	½
Work done in moving q_2 against the field due to q_1	
$$W_3 = \frac{1}{4\pi\epsilon_o}\frac{q_1q_2}{r}$$	½
Hence total work done is $W = W_1 + W_2 + W_3$	
$$W = q_1V_1 + q_2V_2 + \frac{1}{4\pi\epsilon_o}\frac{q_1q_2}{r}$$	½
[V_1 and V_2 are potentials at the two points in the electric field]	
(b)	
(i)	
$$E = \frac{-dV}{dx} = -\frac{d}{dx}(10x + 5)$$	1
$$\therefore \vec{E} = -10\hat{\imath}\ N/C$$	½
(ii) Electric flux through the cube, ϕ =sum of electric flux through 6 faces	
Electric flux through faces perpendicular Y and Z axis = 0	½
$\because$ E is along x axis	
Electric flux through faces perpendicular to x axis	
$= \phi_1 + \phi_2$	½
$= 10 \times (0.2)^2 - 10 \times (0.2)^2$	
=0	½

2. An electric dipole consisting of charges $+q$ and $-q$ separated by a distance L is in stable equilibrium in a uniform electric field $\vec{E}$. The electrostatic potential energy of the dipole is *1*

(A) qLE

(B) zero

(C) – qLE

(D) – 2 qEL

CBSE 20

2. An electric dipole consisting of charges $+q$ and $-q$ separated by a distance L is in stable equilibrium in a uniform electric field $\vec{E}$. The electrostatic potential energy of the dipole is *1*

(A) qLE

(B) zero

(C) – qLE

(D) – 2 qEL

CBSE 20

(C)

$-qLE$

(a) Find the expression for the potential energy of a system of two point charges q_1 and q_2 located at $\vec{r}_1$ and $\vec{r}_2$, respectively in an external electric field $\vec{E}$.

(b) Draw equipotential surfaces due to an isolated point charge (– q) and depict the electric field lines.

(c) Three point charges + 1 μC, – 1 μC and + 2 μC are initially infinite distance apart. Calculate the work done in assembling these charges at the vertices of an equilateral triangle of side 10 cm. *5*

CBSE 20

b)

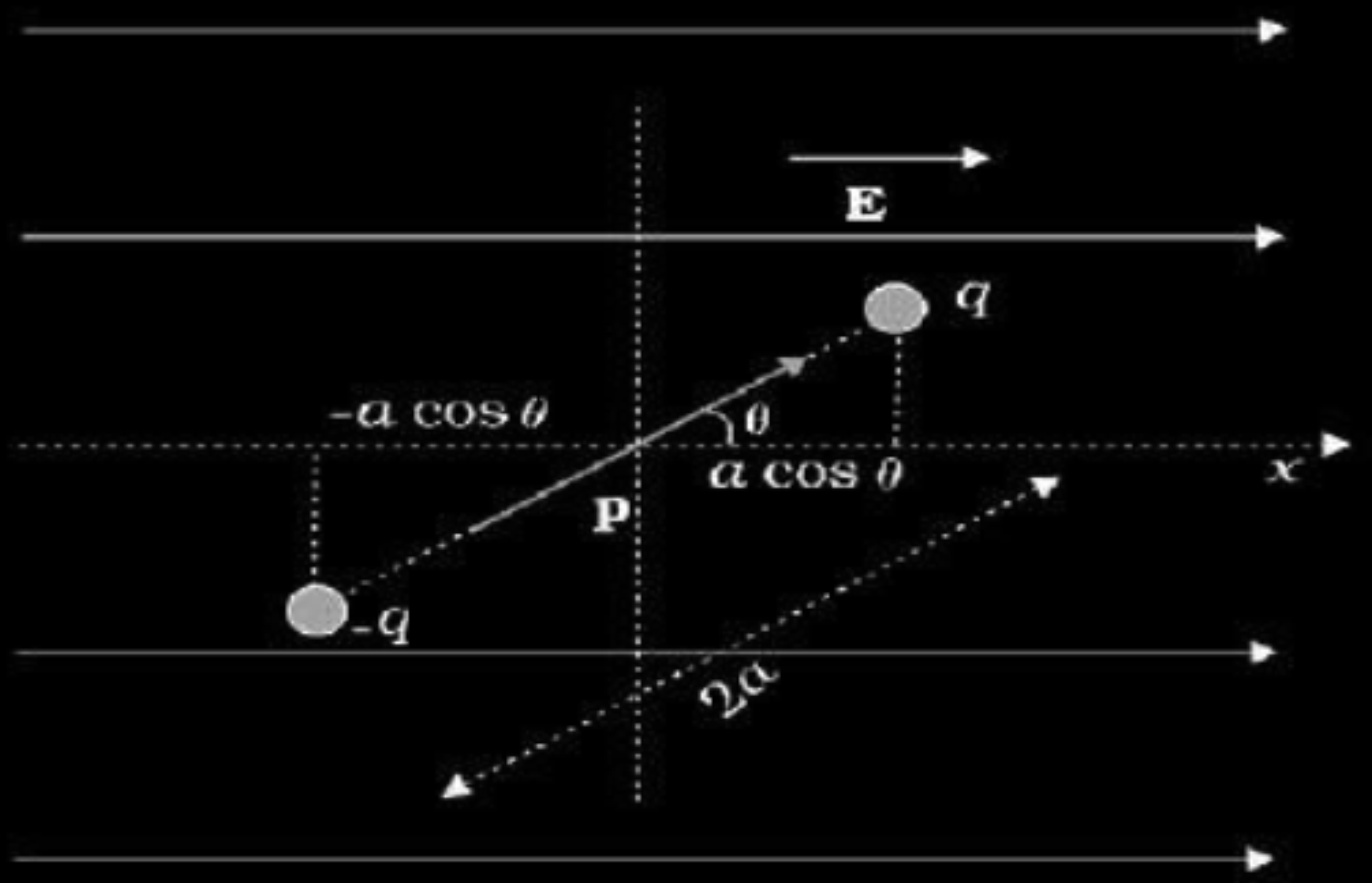
(a) Work done in bringing q from infinity against the field $E = q_1 V \lvert\vec{r_1}\rvert$	1
Work done on q_2 against the field $E = q_2 V \lvert\vec{r_2}\rvert$ Work done on q_2 against the field due to q_1 $= \dfrac{q_1 q_2}{4\pi\epsilon_0 (r_{12})}$	1
Potential energy of the system= Total work done in assembling the system	½
$= q_1 V(\vec{r_1}) + q_2 V(\vec{r_2}) + \dfrac{q_1 q_2}{4\pi\epsilon_0 r_{12}}$	½
c) Work done= charge in potential energy	1
$= \dfrac{kq_1q_2}{r_{12}} + \dfrac{kq_1q_3}{r_{13}} + \dfrac{kq_2q_{31}}{r_{23}}$	½
$= \dfrac{9\times10^{9}\times10^{-12}}{0.1}[1\times -1 + -1\times 2 + 1\times 2]$ $= 9\times10^{-2}[-1-2+2]$ $= -9\times10^{-2} J$	½

(a) Derive an expression for the potential energy of an electric dipole in a uniform electric field. Explain conditions for stable and unstable equilibrium.

(b) Is the electrostatic potential necessarily zero at a point where the electric field is zero ? Give an example to support your answer.

CBSE 19

a) Since torque acting on dipole



$$\vec{\tau} = \vec{p} \times \vec{E}$$ ½

$$\vec{\tau} = pE \sin\theta . \hat{n}$$ ½

work done $d\omega = \tau . d\theta$

$$= pE\sin\theta d\theta$$

$$w = \int_{\theta_1}^{\theta_2} dw \quad pE \int_{\theta_1}^{\theta_2} \sin\theta \, d\theta$$ ½

$$w = pE[-\cos\theta]_{\theta_1}^{\theta_2}$$

$$= pE[\cos\theta_1 - \cos\theta_2]$$

if $\theta_1 = 0, \theta_2 = \theta$

$$w = pE(1-\cos\theta)$$ ½

Conditions-

For stable equilibrium - When electric dipole is parallel to electric field. 1

For unstable equilibrium - Anti Parallel to electric field. 1

b) No. ½

Inside equipotential surface ½

An electron is accelerated through a potential difference V. Write the expression for its final speed, if it was initially at rest. CBSE 18

Two point charges q and –q are located at points (0, 0, –a) and (0, 0, a) respectively.

(a) Find the electrostatic potential at (0, 0, z) and (x, y, 0)

(b) How much work is done in moving a small test charge from the point (5, 0, 0) to (–7, 0, 0) along the x-axis ?

(c) How would your answer change if the path of the test charge between the same points is not along the x-axis but along any other random path ?

(d) If the above point charges are now placed in the same positions in a uniform external electric field $\vec{E}$, what would be the potential energy of the charge system in its orientation of unstable equilibrium ?

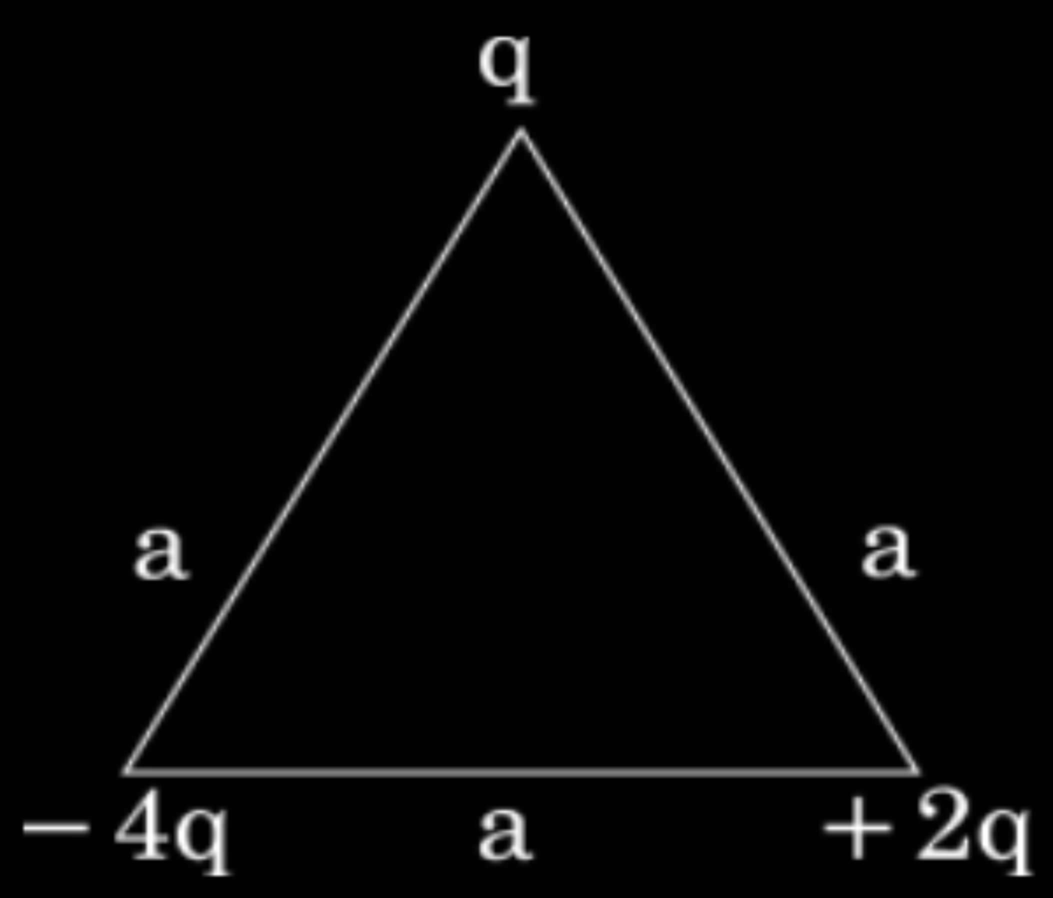
Justify your answer in each case.

CBSE 18

(a) Explain why, for any charge configuration, the equipotential surface through a point is normal to the electric field at that point.

Draw a sketch of equipotential surfaces due to a single charge $(-q)$, depicting the electric field lines due to the charge.

(b) Obtain an expression for the work done to dissociate the system of three charges placed at the vertices of an equilateral triangle of side 'a' as shown below.



CBSE 16

(a) If the field is not normal to an equipotential surface, it would have a non zero component along the surface. This would imply that work would have to be done to move a charge on the surface which is contradictory to the definition of equipotential surface. 1 ½

(Alternatively,

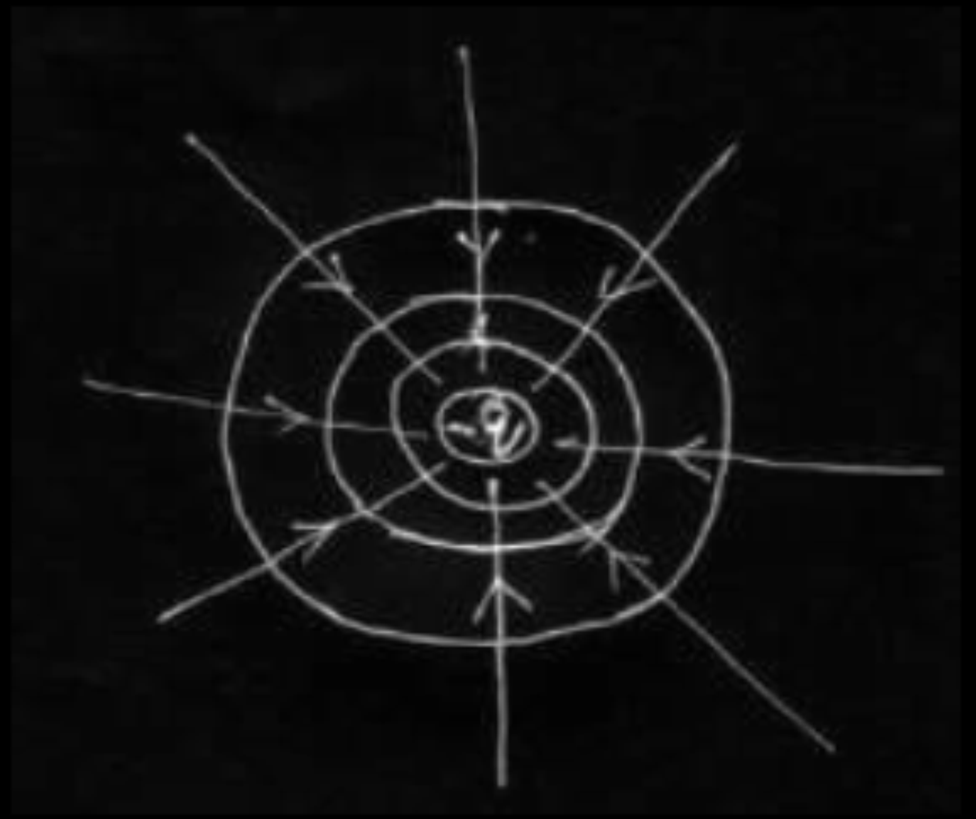
Work done to move a charge dq, on a surface, can be expressed as

$$dW = dq(\vec{E}.\overrightarrow{dr})$$ ½

But $dW=0$ on an equipotential surface ½

$\therefore \vec{E} \perp \overrightarrow{dr})$ ½

Equipotential surfaces for a charge –q



½ +½

(b) Work done to dissociate the system

= -Potential energy of the system ½

$$= \frac{-1}{4\pi\epsilon_0}\left[\frac{(-4q)(q)}{a} + \frac{(2q)(q)}{a} + \frac{(-4q)(2q)}{a}\right]$$ 1

$$= -\frac{1}{4\pi\epsilon_0 a}[-4q^2 + 2q^2 - 8q^2]$$ ½

$$= +\left[\frac{10q^2}{4\pi\epsilon_0 a}\right]$$ ½